# FLIKT.AI

AI-Powered Plan Coordination

# Plan Conflict & Coordination Report

**Project Name:** Meridian Residence

**Report Date:** May 14, 2026

**Analyst:** Flikt.AI Automated Analysis

# Executive Summary

| 44 | 4 | 21 | 19 | 0 |
|---|---|---|---|---|
| Total | Critical | High | Medium | Low |

| $91,206 – $215,882 | 12 |
|---|---|
| Cost Exposure | Discipline Pairs |

## Severity Distribution


## Top Priority Actions

**1. Concrete Block Wall vs Metal Stud Partition Coordination at Second Level**

Severity: Critical | Architectural ↔ Structural

Structural engineer should provide a wall schedule or markup identifying which walls on the second level are structural masonry vs. non-structural partitions. Verify that the 25 P.S.F. partition allowance on S5 adequately covers the architectural wall assemblies specified on A4, including tile and stone finishes in wet areas.

**2. Roof Terrace Parapet and Railing Structural Support Not Detailed**

Severity: Critical | Architectural ↔ Structural

Structural engineer must provide parapet reinforcement details, glass railing post anchorage details with embedment specifications, and connection details that account for the design wind pressures shown on S5. These details should be added to the structural drawing set before permit submission.

**3. Smoke Detector Locations Not Shown on Electrical Plans**

Severity: Critical | Electrical-Lighting ↔ Architectural

Electrical engineer to add smoke detector/alarm locations to all electrical floor plans per FBC Residential and NFPA 72 requirements. Provide interconnected smoke alarms in each bedroom, outside each sleeping area, on each level, and in the understory. Coordinate with architectural plans for exact bedroom locations. Specify hardwired with battery backup per code.

# Detailed Findings

Critical

ID: C001

## Concrete Block Wall vs Metal Stud Partition Coordination at Second Level

| | | | |
|---|---|---|---|
| **Disciplines:** | Architectural ↔ Structural | **Type:** | Missing Coordination |
| **Location:** | Second Level perimeter and interior walls | **Sheets:** | A4, S5, S4, A1, A8 |

**Description:** The architectural legend on A4 describes two wall types: 8" concrete block walls (with 5/8" gypsum wallboard, 1-5/8" metal furring channels, R5 insulation, fire-stop furring channels) for exterior/structural walls, and 20 GA 3-5/8" metal framed interior partitions for interior walls. The structural notes on S5 reference 8" reinforced masonry walls (60 P.S.F.) and 8" masonry walls (60 P.S.F.) at the ground floor slab level, and the Second Level design loads list 'Partition' at 25 P.S.F. However, there is no structural detail or schedule confirming which second-level walls are load-bearing masonry vs. non-load-bearing metal stud, and the partition load of 25 P.S.F. must be verified against the actual wall construction specified architecturally. The 8" CMU exterior walls at the second level need structural coordination for reinforcement, lintel, and bond beam details that are not visible on these sheets.

| Construct. | Cost | Safety | Schedule | Downstrm. |
|---|---|---|---|---|
| **6**/10 | **4**/10 | **7**/10 | **4**/10 | **5**/10 |

**Cost Exposure:** $0 – $0 **Schedule Impact:** 5-7 days

**Recommended Action:** Structural engineer should provide a wall schedule or markup identifying which walls on the second level are structural masonry vs. non-structural partitions. Verify that the 25 P.S.F. partition allowance on S5 adequately covers the architectural wall assemblies specified on A4, including tile and stone finishes in wet areas.

Critical

ID: C002

## Roof Terrace Parapet and Railing Structural Support Not Detailed

| | | | |
|---|---|---|---|
| **Disciplines:** | Architectural ↔ Structural | **Type:** | Missing Coordination |
| **Location:** | Roof terrace perimeter - Second Level | **Sheets:** | A4, S5 |

**Description:** Architectural plan A4 calls out a '42" HIGH APP. REINF. CONC PARAPET - SEE STRUCTURAL DWGS + BLDG SECTION' and a '42" HIGH ALUM. FRAMED-GLASS RAILING SYSTEM - PROVIDE SHOP DRAWINGS FOR REVIEW AND APPROVAL' at the roof terrace edges. The structural notes sheet S5 does not include parapet reinforcement details, glass railing anchorage details, or connection specifications for these elements. The design wind pressure table on S5 shows significant negative pressures at roof edges (up to -51.26 P.S.F. for roofing and -84.25 P.S.F. at corners for walls), which require specific structural detailing for parapet and railing anchorage to resist these loads. The architectural note explicitly references structural drawings for the parapet, but no corresponding detail is provided on S5.

| Construct. | Cost | Safety | Schedule | Downstrm. |
|---|---|---|---|---|
| **6**/10 | **6**/10 | **8**/10 | **5**/10 | **5**/10 |

**Cost Exposure:** $0 – $0 **Schedule Impact:** 7-14 days

**Recommended Action:** Structural engineer must provide parapet reinforcement details, glass railing post anchorage details with embedment specifications, and connection details that account for the design wind pressures shown on S5.

These details should be added to the structural drawing set before permit submission.

## Critical — ID: C003 Smoke Detector Locations Not Shown on Electrical Plans

| | | | |
|---|---|---|---|
| **Disciplines:** | Electrical-Lighting ↔ Architectural | **Type:** | Missing Coordination |
| **Location:** | Throughout residence, all levels | **Sheets:** | E2, E3, E1, A2, A3 |

**Description:** The electrical symbol legend on E3 includes a symbol for 'SMOKE DETECTOR (PHOTOELECTRIC TYPE)' but reviewing the electrical plans E1 (understory), E2 (first level), and E3 (second level), smoke detector locations are not clearly shown on the floor plans. Per Florida Building Code and NFPA 72 for residential construction, smoke detectors/alarms are required in each sleeping room, outside each sleeping area, and on each level of the dwelling. This three-story residence with multiple bedrooms on the first and second levels requires smoke alarms in bedrooms #1, #2, #3, master bedroom, and in hallways/landings outside sleeping areas. The absence of smoke detector locations on the electrical plans is a life-safety coordination gap.

| Construct. | Cost | Safety | Schedule | Downstrm. |
|---|---|---|---|---|
| **2**/10 | **2**/10 | **8**/10 | **2**/10 | **3**/10 |

**Cost Exposure:** $0 – $0 **Schedule Impact:** 2 days

**Recommended Action:** Electrical engineer to add smoke detector/alarm locations to all electrical floor plans per FBC Residential and NFPA 72 requirements. Provide interconnected smoke alarms in each bedroom, outside each sleeping area, on each level, and in the understory. Coordinate with architectural plans for exact bedroom locations. Specify hardwired with battery backup per code.

## Critical — ID: C004 Glass Guardrail at Stair Opening - Load Path Not Defined

| | | | |
|---|---|---|---|
| **Disciplines:** | Architectural ↔ Structural | **Type:** | Missing Coordination |
| **Location:** | Second Level - 'Open to Below' area and landing | **Sheets:** | A4, S5 |

**Description:** The architectural plan A4 shows a '42" HIGH ALUM. FRAMED-GLASS RAILING' at the 'OPEN TO BELOW' area on the second level, which serves as a guardrail at the floor opening. Per IBC/FBC, guardrails at floor openings must resist a 200 lb concentrated load and 50 plf distributed load at the top rail. The structural sheet S5 does not include any details for the guardrail post anchorage into the 8" concrete slab edge, nor does it address the slab edge reinforcement needed to resist the guardrail loads. This is a life-safety element requiring structural engineering.

| Construct. | Cost | Safety | Schedule | Downstrm. |
|---|---|---|---|---|
| **7**/10 | **5**/10 | **9**/10 | **5**/10 | **5**/10 |

**Cost Exposure:** $0 – $0 **Schedule Impact:** 7-10 days

**Recommended Action:** Structural engineer must detail the glass railing post anchorage at the floor opening edge, including slab edge reinforcement, post base plate connection, and verification that the concrete slab edge can resist the code-required guardrail loads. This detail should be provided before permit submission.

High

ID: C011

## HVAC Duct Routing Through Concrete Beam Zone - First Level

| | | | |
|---|---|---|---|
| **Disciplines:** | Mechanical ↔ Structural | **Type:** | Spatial Clash |
| **Location:** | First level ceiling space, kitchen/dining/living areas | **Sheets:** | M3, S1, A1, M2, S3, A9 |

**Description:** (See plan set for details)

| Construct. | Cost | Safety | Schedule | Downstrm. |
|---|---|---|---|---|
| **7**/10 | **5**/10 | **1**/10 | **6**/10 | **7**/10 |

**Cost Exposure:** $11,651 – $27,233 **Schedule Impact:** 12 days

**Recommended Action:** Perform a detailed ceiling plenum coordination study overlaying mechanical duct routing on the structural framing plan. Verify that duct heights plus insulation plus clearance fit below the lowest structural beam soffit while maintaining required finished ceiling heights. Consider reducing duct heights by increasing widths where beam conflicts occur. Provide reflected ceiling plan coordination between architectural, structural, and mechanical.

High

ID: SP037

## HVAC Load Calculations Show Zero Outdoor Air Ventilation for Occupied Spaces

| | | | |
|---|---|---|---|
| **Disciplines:** | Architectural ↔ Specifications | **Type:** | Code Violation |
| **Location:** | Spec Page 10 - Ventilation Sizing Summary for AHU2 | **Sheets:** | Mechanical plans (spec cross-check), Architectural plans (spec cross-check) |

**Description:** The ventilation sizing summary for AHU2 shows a Design Ventilation Airflow Rate of 0 CFM for all 13 spaces on the second floor, including occupied bedrooms (Master Bed, Bedroom 1, Bedroom 2) with listed occupants (2 persons each) and a Gym space. The Required Outdoor Air is listed as 0.0 CFM for every space. For a residential project in Miami, Florida, ASHRAE 62.2 and the Florida Building Code require mechanical ventilation for habitable spaces. Zero outdoor air ventilation for bedrooms and a gym is a code compliance concern that the architectural plans should address but cannot be verified due to missing plan content.

| Construct. | Cost | Safety | Schedule | Downstrm. |
|---|---|---|---|---|
| **6**/10 | **5**/10 | **7**/10 | **4**/10 | **5**/10 |

**Cost Exposure:** $11,203 – $26,175 **Schedule Impact:** 1-2 weeks

**Recommended Action:** Coordinate with the MEP engineer (the design team) to confirm that outdoor air ventilation is being provided by a separate system (e.g., ERV, dedicated OA unit) or revise the AHU2 calculations to include code-required outdoor air rates per ASHRAE 62.2 / Florida Building Code.

**High**

ID: SP034

## Zero Outdoor Air Ventilation for Occupied Residential Spaces

| | | | |
|---|---|---|---|
| **Disciplines:** | Architectural ↔ Specifications | **Type:** | Code Violation |
| **Location:** | Spec Pages 10 - Ventilation Sizing Summary for AHU2 | **Sheets:** | Architectural plans (spec cross-check), Mechanical plans (spec cross-check) |

**Description:** The HVAC load calculation (Spec Page 10) shows a Design Ventilation Airflow Rate of 0 CFM for AHU2, with all spaces including occupied bedrooms (Master Bed, Bedroom 1, Bedroom 2 with listed occupants of 2 each) and a Gym showing 0.0 CFM required outdoor air. For a Miami, Florida location, ASHRAE 62.2 and the Florida Building Code require mechanical ventilation for residential buildings. The architectural plans (signed by Roger Piper, Architect) do not appear to contain any information addressing how outdoor air ventilation will be provided to these occupied spaces, nor is there any note referencing a separate ventilation system or ERV/HRV.

| Construct. | Cost | Safety | Schedule | Downstrm. |
|---|---|---|---|---|
| **6**/10 | **5**/10 | **7**/10 | **4**/10 | **5**/10 |

**Cost Exposure:** $11,025 – $25,966 **Schedule Impact:** 1-2 weeks

**Recommended Action:** Coordinate between the architect and mechanical engineer (the design team) to confirm how code-required ventilation will be provided. Either update the HAP model to include outdoor air requirements per ASHRAE 62.2/Florida Building Code, or document a separate dedicated outdoor air system (DOAS/ERV) on both the architectural and mechanical plans.

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**High**

ID: SP028

## No Duct Layout or Diffuser Schedule to Match Space Airflow Requirements

| | | | |
|---|---|---|---|
| **Disciplines:** | Mechanical ↔ Specifications | **Type:** | Specification Mismatch |
| **Location:** | Spec Page 9 – Space Loads and Airflows vs. Mechanical Plans | **Sheets:** | Mechanical plans (spec cross-check) |

**Description:** The specification provides detailed space-by-space airflow requirements (e.g., Master Bed 436 CFM, Landing/Stairs 344 CFM, Master Bath 319 CFM, Bedroom 1 214 CFM, Gym 176 CFM, etc.) totaling 2,000 CFM across 13 spaces. The mechanical plans contain no duct sizing, diffuser/register schedule, or supply/return air distribution layout to verify these airflows can be delivered to each space.

| Construct. | Cost | Safety | Schedule | Downstrm. |
|---|---|---|---|---|
| **6**/10 | **5**/10 | **7**/10 | **4**/10 | **5**/10 |

**Cost Exposure:** $9,567 – $21,885 **Schedule Impact:** 1-2 weeks

**Recommended Action:** Issue mechanical plans with complete duct layout showing duct sizes, diffuser/grille locations and sizes, and a diffuser schedule that corresponds to the calculated space airflow rates from the HAP analysis.

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## High — ID: SP025

## Mechanical Plans Entirely Absent – No Equipment or Ductwork Shown

| | | | |
|---|---|---|---|
| **Disciplines:** | Mechanical ↔ Specifications | **Type:** | Specification Mismatch |
| **Location:** | All Mechanical Plan Sheets | **Sheets:** | Mechanical plans (spec cross-check) |

**Description:** The specification book provides detailed sizing data for AHU2, a split air handling unit serving 1,514 sq ft across 13 spaces on the second floor (bedrooms, bathrooms, gym, laundry, master suite, etc.) with 2,000 CFM design airflow and 3.7 tons of cooling capacity. However, the plan sheets contain only digital signature blocks from Edward J Kranz dated 2023.10.27 with no discernible mechanical drawings, equipment schedules, duct layouts, or any technical content whatsoever.

| Construct. | Cost | Safety | Schedule | Downstrm. |
|---|---|---|---|---|
| **6**/10 | **5**/10 | **7**/10 | **4**/10 | **5**/10 |

**Cost Exposure:** $9,294 – $21,565 **Schedule Impact:** 1-2 weeks

**Recommended Action:** Obtain and issue complete mechanical plan sheets showing AHU2 location, ductwork routing, diffuser/register locations, refrigerant line routing, condensing unit placement, and all associated mechanical details required to construct the system described in the specifications.

## High — ID: SP031

## AHU2 Equipment Location and Condensing Unit Placement Not Shown

| | | | |
|---|---|---|---|
| **Disciplines:** | Mechanical ↔ Specifications | **Type:** | Specification Mismatch |
| **Location:** | Spec Page 8 – Air System Information (SPLT AHU) vs. Mechanical Plans | **Sheets:** | Mechanical plans (spec cross-check) |

**Description:** The specifications identify AHU2 as a split air handling unit (SPLT AHU) with 3.7 tons cooling and 23.5 MBH heating capacity. A split system requires both an indoor air handler and an outdoor condensing unit with refrigerant piping between them. The mechanical plans show no equipment locations, clearance zones, structural support requirements, refrigerant line routing, or electrical connection points for either component.

| Construct. | Cost | Safety | Schedule | Downstrm. |
|---|---|---|---|---|
| **6**/10 | **5**/10 | **7**/10 | **4**/10 | **5**/10 |

**Cost Exposure:** $9,294 – $21,565 **Schedule Impact:** 1-2 weeks

**Recommended Action:** Provide mechanical plans showing the indoor air handler location with required service clearances, outdoor condensing unit location with manufacturer-required clearances, refrigerant line routing, condensate drain routing, and coordinate with electrical plans for power connections and with structural for equipment support.

High

ID: C012

## Augercast Pile Capacity and Lateral Load Design vs. Design Wind Pressure

| | | | |
|---|---|---|---|
| **Disciplines:** | Structural ↔ Structural | **Type:** | Specification Mismatch |
| **Location:** | Foundation — entire building | **Sheets:** | A4, S5 |

**Description:** The structural notes on S5 specify 14" diameter augercast piles with 'lateral capacity with 4 #6 longitudinal bars & a #3 @ 5" at top (3'-0") & 18" o.c. below.' The design wind pressure table shows significant uplift forces: Roofing Zone 1 at -87.26/-87.26 PSF gross uplift, and Wall Zone 4 (Field Negative) at -34.27 PSF, Zone 5 (Corner Negative) at -44.79 PSF. The pile depth is specified as 45 ft minimum 'as determined by soil boring report and geotechnical engineer's recommendations, from August 1, 2023 by AMERICAN TESTING MATERIALS.' However, the pile lateral capacity design references are incomplete — the notes state piles must 'attain a 30 day breaking strength of 5000 P.S.I.' but do not specify the required axial capacity (tons) per pile or the net uplift resistance per pile. With high wind uplift forces in a Surfside, Florida coastal location (HVHZ), the pile-to-grade-beam connection for uplift resistance must be explicitly detailed.

| Construct. | Cost | Safety | Schedule | Downstrm. |
|---|---|---|---|---|
| 4/10 | 5/10 | 7/10 | 4/10 | 5/10 |

**Cost Exposure:** $5,534 – $12,780 **Schedule Impact:** 7-10 days

**Recommended Action:** Verify pile axial capacity (compression and tension) is specified on the foundation plan. Confirm pile-to-grade-beam connection details include uplift reinforcement adequate for the calculated net uplift forces. Cross-reference geotechnical report recommendations for pile capacity with structural calculations. Ensure Miami-Dade HVHZ requirements are met for foundation anchorage.

High

ID: C005

## MEP Systems Not Shown - No Mechanical, Electrical, or Plumbing Drawings Provided

| | | | |
|---|---|---|---|
| **Disciplines:** | Architectural ↔ Structural | **Type:** | Documentation Gap |
| **Location:** | Entire building - Second Level | **Sheets:** | A4, S5 |

**Description:** The architectural second level floor plan (A4) shows a complex custom residence with Master Bedroom, Master Bath, Gym, Laundry, two additional Bedrooms, multiple bathrooms, and a W.I.C. — all requiring HVAC, plumbing, electrical, and fire protection systems. The structural notes on S5 explicitly state: 'Coordination of the work of all trades (Architectural, Plumbing, Electrical, Air Conditioning and Refrigeration) at the time of erecting the building structure is the responsibility of the contractor.' However, no MEP discipline drawings are provided in this set. Without MEP drawings, there is no verification of: electrical panel sizing and circuit allocation, plumbing riser and fixture connections, HVAC duct routing and equipment sizing, fire sprinkler coverage, or condensate drain routing. The bathroom legend on A4 references fixtures (two-piece flush-tank 1.28 gallon water closet, free-standing cast iron bathtub, shower stall with frameless tempered glass) that require plumbing rough-in coordination not shown.

| Construct. | Cost | Safety | Schedule | Downstrm. |
|---|---|---|---|---|
| 7/10 | 6/10 | 4/10 | 7/10 | 8/10 |

**Cost Exposure:** $0 – $0 **Schedule Impact:** 30-60 days

**Recommended Action:** MEP drawings must be developed and coordinated with the architectural and structural documents before construction. Specifically: verify HVAC equipment locations and structural support, plumbing penetration locations through concrete slabs and CMU walls, electrical service entrance and panel locations, and fire protection system layout. The 2-5 P.S.F. MEP load allowance on S5 should be verified against actual MEP equipment weights.

## High — ID: C006 — Spiral Staircase Delegated Design - Missing Structural Interface Details

| | | | |
|---|---|---|---|
| **Disciplines:** | Architectural ↔ Structural | **Type:** | Missing Coordination |
| **Location:** | Spiral staircase from first level to roof terrace | **Sheets:** | S4, S1, S5, A4, A3, A8, A1, A7 |

**Description:** The architectural floor plan (A3) shows a spiral staircase with detailed notes including 'ALUM. SPIRAL STAIRCASE PROVIDE SHOP DRAWINGS PRIOR TO FABRICATION' and specific dimensional requirements. The architectural elevations (A7) show the spiral staircase on both south and north elevations. The structural framing plans (S1, S1) and section S4 all note 'STAIRS BY OTHERS PROVIDE SIGNED & SEALED SHOP DWGS FOR REVIEW & APPROVAL BY E.O.R.' This delegation is noted but the structural drawings do not show: (a) the slab opening/penetration dimensions for the spiral stair at each level, (b) reinforcement details around the stair opening, (c) connection details for the stair to the concrete slab at each landing, or (d) load transfer requirements. The architectural notes on A3 specify 'CLEAR STAIR ENCLOSURE 5'-0" MIN. MEASURED AT AND BELOW THE HANDRAIL' and specific tread/riser dimensions, but the structural interface is undefined.

| Construct. | Cost | Safety | Schedule | Downstrm. |
|---|---|---|---|---|
| **5**/10 | **5**/10 | **6**/10 | **4**/10 | **6**/10 |

**Cost Exposure:** $0 – $0 **Schedule Impact:** 14 days

**Recommended Action:** Structural engineer to provide: (1) slab opening dimensions and edge reinforcement details at each floor level for the spiral stair penetration, (2) connection requirements (embed plates, anchor bolts) for the stair manufacturer, (3) load criteria (dead load, live load, lateral) for the stair design. Coordinate with stair manufacturer early in construction to avoid delays.

## High — ID: C007 — Generator Electrical Connection vs Structural Roof Screen Coordination

| | | | |
|---|---|---|---|
| **Disciplines:** | Electrical-Lighting ↔ Structural | **Type:** | Missing Coordination |
| **Location:** | Roof level | **Sheets:** | E1, E4, S4, A1, A4, S5 |

**Description:** The electrical riser diagram on E1 shows a 'FUTURE GENERATOR' with a 2P-200A disconnect at the roof level, connected to the main electrical system. The generator schedule shows a Generac RG4 rated at 48,000 watts, 144A at 120/240V, natural gas fuel, dimensions 8.1' x 3.05' x 48". The structural sections (S4, S4) show 'GENERATOR' labeled on the roof with a note 'SCREEN BY OTHERS PROVIDE SIGNED & SEALED SHOP DWGS FOR REVIEW & APPROVAL BY E.O.R.' The architectural section A1 also shows the generator behind a mechanical equipment screen. However, there is no structural detail for the generator pad/support on the roof slab, no vibration isolation detail, and no coordination between the generator weight (approximately 1,800+ lbs for this model) and the roof structural capacity. The structural drawings delegate the screen to 'others' but do not address the generator mounting.

| Construct. | Cost | Safety | Schedule | Downstrm. |
|---|---|---|---|---|
| 5/10 | 5/10 | 5/10 | 4/10 | 6/10 |

**Cost Exposure:** $0 – $0 **Schedule Impact:** 7 days

**Recommended Action:** Structural engineer to provide a generator equipment pad detail on the roof slab, including point load analysis for the generator weight. Coordinate vibration isolation requirements per mechanical schedule with structural support. Provide conduit routing from generator to main electrical panel through the roof structure.

High

ID: SP025

## Mechanical Plans Entirely Missing – No Equipment, Ductwork, or Layout Shown

| | | | |
|---|---|---|---|
| **Disciplines:** | Mechanical ↔ Specifications | **Type:** | Missing Coordination |
| **Location:** | All Mechanical Plan Sheets | **Sheets:** | Mechanical plans (spec cross-check) |

**Description:** The specification book provides detailed sizing data for AHU2 (a split AHU, SZCAV system serving 1,514 ft² across 13 spaces on the second floor) including 2,000 CFM airflow, 3.7-ton cooling coil, 23.5 MBH heating coil, and complete zone-by-zone airflow breakdowns. However, the plan sheets contain only digital signature blocks from Edward J Kranz dated 2023.10.27 with no discernible mechanical drawings, equipment schedules, duct layouts, or any technical content whatsoever.

| Construct. | Cost | Safety | Schedule | Downstrm. |
|---|---|---|---|---|
| 6/10 | 5/10 | 6/10 | 4/10 | 5/10 |

**Cost Exposure:** $0 – $0 **Schedule Impact:** 1-2 weeks

**Recommended Action:** Obtain and issue complete mechanical plan sheets showing AHU2 location, ductwork routing, diffuser/register locations, refrigerant line routing, condensing unit placement, and all associated mechanical details matching the specification sizing data.

High

ID: C008

## Trellis and Equipment Screen - Delegated Design Without Structural Connection Details

| | | | |
|---|---|---|---|
| **Disciplines:** | Architectural ↔ Structural | **Type:** | Missing Coordination |
| **Location:** | Roof level parapet area | **Sheets:** | A4, S5, S4, A1, A8 |

**Description:** The architectural sections (A8, A1) show '4'-6" ALUM FRAMED TRELLIS SUBMIT SHOP DWGS TO ARCHITECT FOR REVIEW & APPROVAL' suspended from the parapet. The structural sections (S4, S4) note 'TRELLIS BY OTHERS PROVIDE SIGNED & SEALED SHOP DWGS FOR REVIEW & APPROVAL BY E.O.R.' Similarly, the mechanical equipment screen is noted as 'SCREEN BY OTHERS PROVIDE SIGNED & SEALED SHOP DWGS.' Neither the architectural nor structural drawings provide: (a) connection details to the concrete parapet, (b) wind load requirements for the trellis/screen design (critical in Surfside, FL coastal zone), (c) embed plate or anchor locations in the parapet. The wind load calculations (C&C; sheet) show significant wind pressures (-87 to -157 psf on roof zones), making the trellis and screen attachment design critical.

| Construct. | Cost | Safety | Schedule | Downstrm. |
|---|---|---|---|---|
| 6/10 | 7/10 | 5/10 | 2/10 | 7/10 |

**Cost Exposure:** $0 – $0 **Schedule Impact:** 12 days

**Recommended Action:** Structural engineer to provide: (1) embed plate or post-installed anchor requirements in the concrete parapet for trellis and screen attachment, (2) design wind loads for the trellis and screen based on the C&C; wind load analysis, (3) connection capacity requirements for the shop drawing submittals. Include these requirements in the structural general notes or a dedicated detail sheet.

High

ID: SP036

## Architectural Plan Sheets Contain No Architectural Design Information

| | | | |
|---|---|---|---|
| **Disciplines:** | Architectural ↔ Specifications | **Type:** | Documentation Gap |
| **Location:** | All Plan Sheets (the architect signed sheets) | **Sheets:** | Architectural plans (spec cross-check) |

**Description:** The architectural plan sheets extracted contain only digital signature blocks from the architect Inc. and structural calculation output from spBeam (the design team). No actual architectural floor plans, room layouts, dimensions, wall sections, or schedules are present in the plan sheet text. The specification book (HVAC load calculations by the design team) references 13 distinct second-floor spaces (Master Bed, Master Bath, Bedrooms 1 & 2, Gym, Laundry, etc.) with specific floor areas totaling 1,514 SF, but there is no corresponding architectural plan documentation to verify room sizes, layouts, or configurations.

| Construct. | Cost | Safety | Schedule | Downstrm. |
|---|---|---|---|---|
| 6/10 | 5/10 | 6/10 | 4/10 | 5/10 |

**Cost Exposure:** $0 – $0 **Schedule Impact:** 1-2 weeks

**Recommended Action:** Obtain and review the actual architectural plan sheets (floor plans, elevations, sections, schedules) for the the architect Residence to verify room areas, layouts, and coordination with the HVAC load calculations. The current plan set appears to be incomplete or improperly extracted.

High

ID: C009

## Terrace Stone Veneer and Railing Loads Not Explicitly Addressed in Roof Terrace Design Load

| | | | |
|---|---|---|---|
| **Disciplines:** | Architectural ↔ Structural | **Type:** | Missing Coordination |
| **Location:** | Second Level Terraces (East and West) | **Sheets:** | A4, S5 |

**Description:** Architectural plan A4 shows terraces with 2" thick stone veneer over cement board and mesh fastened to structural steel beam, 42" high aluminum/glass railings, 42" high reinforced concrete parapets, and porcelain tile flooring on concrete deck. The structural design load on S5 for 'Roof Terrace' lists 8" concrete slab at 100 P.S.F., insulation at 10 P.S.F., tile at 10 P.S.F., ceiling at 5 P.S.F., and Mech/Elect/Misc at 3 P.S.F. for a total D.L. of 128 P.S.F. However, the stone veneer cladding on terrace walls, the concrete parapet weight, the decorative aluminum powder-coat screen, and the frameless glass railing system are not explicitly itemized in the design load table. These elements add significant point and line loads to the terrace structure.

| Construct. | Cost | Safety | Schedule | Downstrm. |
|---|---|---|---|---|
| 5/10 | 5/10 | 6/10 | 4/10 | 6/10 |

**Cost Exposure:** $0 – $0 **Schedule Impact:** 5-10 days

**Recommended Action:** (See plan set for details)

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**High**

ID: C010

## Gas Piping to Generator Not Shown on Gas Isometric

| | | | |
|---|---|---|---|
| **Disciplines:** | Plumbing ↔ Mechanical | **Type:** | Missing Coordination |
| **Location:** | Roof level, generator location | **Sheets:** | P6, E1, M4 |

**Description:** The generator schedule on E1 specifies a Generac RG4 fueled by natural gas with fuel consumption of 334 CF/HR at full load and 204 CF/HR at half load. The gas isometric on P6 shows gas piping routing to various appliances (range, water heater, dryer, etc.) but does not appear to include a gas line routed to the roof level for the generator. The total gas load noted on P6 is 997.20 MBH, but it is unclear if this includes the generator demand. The generator at 48kW would add approximately 400-500 MBH of gas demand, which would significantly impact the gas service sizing.

| Construct. | Cost | Safety | Schedule | Downstrm. |
|---|---|---|---|---|
| 5/10 | 6/10 | 7/10 | 5/10 | 7/10 |

**Cost Exposure:** $0 – $0 **Schedule Impact:** 5-7 days

**Recommended Action:** Plumbing engineer to add gas piping to the roof-level generator on the gas isometric (P6). Recalculate total gas load including generator demand and verify gas meter and service line sizing is adequate. Coordinate routing with structural for roof penetration.

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**High**

ID: SP035

## Architectural Plans Lack Room Layout and Dimensional Data to Verify HVAC Load Calculations

| | | | |
|---|---|---|---|
| **Disciplines:** | Architectural ↔ Specifications | **Type:** | Documentation Gap |
| **Location:** | Spec Pages 9-11 (AHU2 Zone Sizing) vs. Architectural Plan Sheets | **Sheets:** | Architectural plans (spec cross-check), Mechanical plans (spec cross-check) |

**Description:** The specification/load calculation documents list 13 specific second-floor spaces with precise floor areas (e.g., Master Bed 216 ft², Master Bath 165 ft², Gym 130 ft², Laundry 95 ft², etc.) totaling 1,514 ft² served by AHU2. The architectural plan sheets provided contain only digital signature blocks from Roger Piper and structural calculation data from the design team — no architectural floor plans, room schedules, or dimensional information is legible or present to verify that the room areas and program match the architectural drawings.

| Construct. | Cost | Safety | Schedule | Downstrm. |
|---|---|---|---|---|
| 6/10 | 5/10 | 5/10 | 4/10 | 5/10 |

**Cost Exposure:** $0 – $0 **Schedule Impact:** 3-5 days

**Recommended Action:** Obtain and cross-reference the actual architectural floor plan sheets (A-series) to verify that room names, areas, and configurations match the HVAC load calculation assumptions. Confirm the second-floor area of 1,514 ft² and individual space areas align with the architectural drawings.

High

ID: SP028

## Duct Layout and Diffuser Locations Not Coordinated with Space Load Data

| | | | |
|---|---|---|---|
| **Disciplines:** | Mechanical ↔ Specifications | **Type:** | Missing Coordination |
| **Location:** | Spec Page 9 – Space Loads and Airflows vs. Mechanical Plans | **Sheets:** | Mechanical plans (spec cross-check) |

**Description:** The specifications provide individual space-level airflow requirements for 13 spaces (e.g., Master Bed at 436 CFM, Landing/Stairs at 344 CFM, Master Bath at 319 CFM, Bedroom 1 at 214 CFM, Gym at 176 CFM, etc.). The mechanical plans contain no duct routing, branch sizing, diffuser schedules, or register locations to demonstrate these airflows will be delivered to each space as designed.

| Construct. | Cost | Safety | Schedule | Downstrm. |
|---|---|---|---|---|
| **6**/10 | **5**/10 | **7**/10 | **4**/10 | **5**/10 |

**Cost Exposure:** $0 – $0 **Schedule Impact:** 1-2 weeks

**Recommended Action:** Issue mechanical plans with complete duct layout showing branch duct sizes, diffuser/register types and sizes, and CFM annotations for each space matching the specification sizing summary. Include a duct sizing schedule and ensure coordination with architectural floor plans.

High

ID: C013

## Electrical Panel Location vs Architectural Wall Layout - Understory

| | | | |
|---|---|---|---|
| **Disciplines:** | Electrical-Lighting ↔ Architectural | **Type:** | Missing Coordination |
| **Location:** | Understory level, storage/utility area | **Sheets:** | E1, A2, E2 |

**Description:** The electrical understory plan (E1) shows Panel 'K' and Panel 'W' locations, along with the meter and electrical service entrance, at the understory level near the vestibule/carport area. The electrical riser diagram shows these panels with main breakers (200A each). The architectural understory plan (A2) shows the storage room and vestibule area but does not specifically call out the electrical panel locations or provide the required NEC clearances (36" front clearance, 30" wide, 78" headroom per NEC 110.26). The understory has a relatively low ceiling height - the architectural sections show the understory floor-to-ceiling is approximately 4'-5" to the underside of the first floor slab in some areas (per section details showing understory height). This raises concerns about the 78" headroom requirement for electrical panels.

| Construct. | Cost | Safety | Schedule | Downstrm. |
|---|---|---|---|---|
| **4**/10 | **3**/10 | **6**/10 | **3**/10 | **4**/10 |

**Cost Exposure:** $0 – $0 **Schedule Impact:** 7 days

**Recommended Action:** Verify that electrical panel locations have adequate NEC 110.26 clearances: 36" front, 30" wide, 78" headroom. Cross-reference panel locations with architectural understory plan to confirm wall locations and ceiling heights are adequate. If understory ceiling height is less than 78" at panel locations, panels must be relocated to a compliant area.

High

ID: SP031

## Heating System Fuel/Type Not Defined on Plans

| | | | |
|---|---|---|---|
| **Disciplines:** | Mechanical ↔ Specifications | **Type:** | Documentation Gap |
| **Location:** | Spec Page 8 – Central Heating Coil Sizing Data | **Sheets:** | Mechanical plans (spec cross-check) |

**Description:** The specifications show a heating coil load of 23.5 MBH with entering DB of 67.4°F and leaving DB of 78.3°F, and water flow listed as N/A, suggesting electric heat or heat pump operation rather than hydronic. However, the heating source type (electric resistance, heat pump, gas) is not explicitly stated in the available specification pages, and the mechanical plans provide no information about the heating system type, fuel source, or associated infrastructure.

| Construct. | Cost | Safety | Schedule | Downstrm. |
|---|---|---|---|---|
| **6**/10 | **5**/10 | **5**/10 | **4**/10 | **5**/10 |

**Cost Exposure:** $0 – $0 **Schedule Impact:** 3-5 days

**Recommended Action:** Clarify the heating source type in both specifications and on mechanical plans. If heat pump, show defrost controls and auxiliary heat. If electric resistance, coordinate electrical panel capacity. Ensure the selected heating method is reflected in the energy code compliance documentation.

High

ID: SP037

## Missing Architectural Plan Content — No Visible Floor Plans, Sections, or Details

| | | | |
|---|---|---|---|
| **Disciplines:** | Architectural ↔ Specifications | **Type:** | Missing Coordination |
| **Location:** | All Architectural Plan Sheets | **Sheets:** | Architectural plans (spec cross-check) |

**Description:** The HVAC specifications (the design team, dated 09/18/2023) define a detailed second-floor program with 13 spaces, 849 ft² of window/skylight area, and 1,152 ft² of wall area for AHU2 Zone 1. However, the architectural plan sheet text contains only repeated digital signatures and embedded structural beam calculations — no architectural drawings, wall sections, window schedules, or construction details are present to confirm or coordinate with the mechanical design assumptions.

| Construct. | Cost | Safety | Schedule | Downstrm. |
|---|---|---|---|---|
| **6**/10 | **5**/10 | **7**/10 | **4**/10 | **5**/10 |

**Cost Exposure:** $0 – $0 **Schedule Impact:** 1-2 weeks

**Recommended Action:** Ensure complete architectural drawing sets (floor plans, elevations, wall sections, window/door schedules) are included in the construction document package. Cross-reference window areas (849 ft²) and wall areas (1,152 ft²) from the HVAC calcs against the architectural elevations and window schedule.

**High**

ID: C014

## Augercast Pile Foundation - No Pile Layout Plan Coordination with Architectural Floor Plan

| | | | |
|---|---|---|---|
| **Disciplines:** | Structural ↔ Architectural | **Type:** | Documentation Gap |
| **Location:** | Foundation level - entire building | **Sheets:** | S5, A4 |

**Description:** Structural notes on S5 specify 'Structure to be supported by 14" diameter augercast piles' with specific requirements: 'Pile depth of 45 ft. min. as determined by soil boring report and geotechnical engineer's recommendations from August 1, 2023 by AMERICAN TESTING MATERIALS.' The notes specify 'Each pile shall have a minimum bearing capacity of 40 tons loaded by a static load test' with '4 #6 longitudinal bars & a #3 @ 5" at top (3'-0") & 18" o.c. balance.' The Ground Floor Slab design load shows 8" Concrete Slab (100 PSF) and 8" Reinforced Masonry Wall (60 PSF and 80 PSF). However, no pile layout plan is provided in the available sheets to verify pile locations coordinate with the architectural layout, particularly at the spiral staircase, elevator (if any), and heavy load points like the stone-clad steel beams.

| Construct. | Cost | Safety | Schedule | Downstrm. |
|---|---|---|---|---|
| **7**/10 | **6**/10 | **7**/10 | **6**/10 | **8**/10 |

**Cost Exposure:** $0 – $0 **Schedule Impact:** 10-14 days

**Recommended Action:** Obtain and review structural pile layout plan to verify pile locations align with architectural load-bearing walls, columns, and concentrated load points. Verify pile capacity is adequate for the cumulative loads including stone veneer, concrete parapets, and rooftop equipment.

**Medium**

ID: C021

## Condensate Drain Routing Through Ceiling Space Conflicts with Concrete Slab Structure

| | | | |
|---|---|---|---|
| **Disciplines:** | Mechanical ↔ Structural | **Type:** | Spatial Clash |
| **Location:** | Second level, AHU2 area near bar/hallway | **Sheets:** | M3, S1, A1 |

**Description:** The second level mechanical plan (M3) shows a note 'ROUTE CONDENSATE DRAIN FOR AHU2 IN CEILING SPACE. PROVIDE CONDENSATE DRAIN PUMP.' This indicates the condensate drain must be routed through the ceiling plenum space below the second floor. However, the structural drawings (S1) show the second floor is an 8-inch concrete slab with concrete beams (2B series). The building uses reinforced concrete slab construction throughout, not open-web joists, meaning the available ceiling plenum space between the underside of the concrete slab and the finished ceiling below is extremely limited. The architectural sections (A1) show the first floor ceiling heights are tight with concrete beams dropping below the slab. Routing condensate drains around concrete beams requires careful coordination to maintain proper slope.

| Construct. | Cost | Safety | Schedule | Downstrm. |
|---|---|---|---|---|
| **6**/10 | **4**/10 | **2**/10 | **5**/10 | **5**/10 |

**Cost Exposure:** $3,584 – $8,724 **Schedule Impact:** 5 days

**Recommended Action:** Coordinate the condensate drain routing with the structural engineer to identify beam locations and available clearances. Verify that the condensate drain pump location and discharge point are feasible within the available plenum space. Provide a routing plan showing how the drain navigates around structural beams while maintaining minimum slope.

## Medium — ID: C022

## Smoke Detector Locations Not Shown on Electrical Plans

| | | | |
|---|---|---|---|
| **Disciplines:** | Electrical-Lighting ↔ Architectural | **Type:** | Code Violation |
| **Location:** | All levels | **Sheets:** | E1, E2, E3, A2, A3 |

**Description:** The electrical symbol legend on E3 includes a symbol for 'SMOKE DETECTOR (PHOTOELECTRIC TYPE)' but reviewing the electrical floor plans (E1 understory, E2 first level, E3 second level), smoke detectors are not clearly shown in all required locations per Florida Building Code and NFPA 72 for a residential occupancy. Per FBC Residential, smoke alarms are required in each sleeping room, outside each sleeping area, and on each level. The residence has multiple bedrooms on the first and second floors, plus the understory level. The electrical plans should show smoke detector locations coordinated with the architectural room layout.

| Construct. | Cost | Safety | Schedule | Downstrm. |
|---|---|---|---|---|
| **2**/10 | **2**/10 | **8**/10 | **2**/10 | **4**/10 |

**Cost Exposure:** $3,082 – $8,114 **Schedule Impact:** 2 days

**Recommended Action:** Add smoke detector locations to all electrical floor plans per FBC and NFPA 72 requirements. Ensure detectors are shown in each bedroom (Bedrooms 1, 2, 3, Master Bedroom), outside each sleeping area in the hallway/landing, and on each level including the understory. Coordinate locations with the architectural ceiling plan to avoid conflicts with lighting fixtures and HVAC diffusers.

## Medium — ID: CV2024

## No fire protection sheets — NFPA 72 fire alarm requirements may not be addressed

| | | | |
|---|---|---|---|
| **Disciplines:** | Fire Protection ↔ Electrical | **Type:** | Code Violation |
| **Location:** | Project-level discipline coverage | **Sheets:** | E4, M3, P3, E1, P4, P5, P2, P6, E3, M4 |

**Description:** Project includes MEP disciplines but no dedicated fire protection sheets. NFPA 72 requires fire alarm and notification appliance documentation. Verify fire protection design is included in the plan set or is being provided under a separate permit.

| Construct. | Cost | Safety | Schedule | Downstrm. |
|---|---|---|---|---|
| **3**/10 | **4**/10 | **6**/10 | **2**/10 | **3**/10 |

**Cost Exposure:** $3,239 – $8,114 **Schedule Impact:** 3-7 days

**Recommended Action:** Confirm fire protection sheets are included in the complete plan set, or verify fire protection is under a separate permit with its own plan review.

**Medium**

ID: SP036

## No Architectural Plan Details for AHU2 Equipment Location or Clearances

| | | | |
|---|---|---|---|
| **Disciplines:** | Architectural ↔ Specifications | **Type:** | Specification Mismatch |
| **Location:** | Spec Page 9 (AHU2 System Info) vs. Architectural Plan Sheets | **Sheets:** | Architectural plans (spec cross-check) |

**Description:** The specification documents define AHU2 as a split air handling unit (SPLT AHU) serving 1,514 ft² of second-floor space with 2,000 CFM design airflow. The architectural plan sheets do not contain any visible equipment location, mechanical room designation, equipment pad, or clearance dimensions for this unit. For a 2,000 CFM AHU, adequate space for installation, service access, and code-required clearances must be coordinated on the architectural plans.

| Construct. | Cost | Safety | Schedule | Downstrm. |
|---|---|---|---|---|
| **6**/10 | **5**/10 | **4**/10 | **4**/10 | **5**/10 |

**Cost Exposure:** $2,868 – $6,828 **Schedule Impact:** 3-5 days

**Recommended Action:** Coordinate with the MEP engineer to establish the AHU2 indoor and outdoor unit locations. The architect should show equipment locations, required clearances, and any structural support or acoustic considerations on the architectural floor plans and/or roof plan.

**Medium**

ID: C020

## Structural Design Load Ceiling Allowance vs Architectural Ceiling Heights

| | | | |
|---|---|---|---|
| **Disciplines:** | Structural ↔ Architectural | **Type:** | Specification Mismatch |
| **Location:** | Second Level rooms and corridors | **Sheets:** | A4, S5 |

**Description:** The structural design load table on S5 lists Second Level dead loads including 8" Concrete Slab (100 PSF), Floor Tile (10 PSF), Partition (25 PSF), Ceiling (3 PSF), and Mech/Elect/Miscl. (2 PSF). The 8" concrete slab with an additional ceiling system below implies a specific floor-to-floor height requirement. The architectural plan A4 shows interior partitions noted as '20 GA. 3-5/8" Metal Framed Interior Partition, Install Stud Spacing at 16" O.C.' The ceiling allowance of only 3 PSF on the second level and 2 PSF for Mech/Elect/Miscl. suggests minimal plenum space. Without building sections showing floor-to-floor heights, it cannot be confirmed that adequate clearance exists between the structural slab soffit and the architectural finish ceiling for HVAC ductwork, sprinkler piping, and lighting fixtures — particularly in the bathrooms where the legend calls for recessed elements.

| Construct. | Cost | Safety | Schedule | Downstrm. |
|---|---|---|---|---|
| **4**/10 | **4**/10 | **4**/10 | **3**/10 | **5**/10 |

**Cost Exposure:** $2,730 – $6,776 **Schedule Impact:** 3-5 days

**Recommended Action:** Architectural drawings should include finished ceiling heights for all second-level rooms. Structural drawings should note bottom-of-structure elevations at each level. Cross-reference these dimensions to confirm adequate plenum space for MEP systems.

Medium

ID: SP032

## Refrigerant Piping and Line Set Routing Not Documented

| | | | |
|---|---|---|---|
| **Disciplines:** | Mechanical ↔ Specifications | **Type:** | Specification Mismatch |
| **Location:** | Spec Page 8 (SPLT AHU designation); Mechanical Plan Sheets | **Sheets:** | Mechanical plans (spec cross-check) |

**Description:** The specification identifies AHU2 as a split system (SPLT AHU), which requires refrigerant line sets between the indoor air handler and outdoor condensing unit. No plan sheets show refrigerant line routing, line sizes, insulation requirements, or maximum allowable line lengths per manufacturer specifications. This is critical for proper system performance and refrigerant charge calculations.

| Construct. | Cost | Safety | Schedule | Downstrm. |
|---|---|---|---|---|
| **6**/10 | **5**/10 | **4**/10 | **4**/10 | **5**/10 |

**Cost Exposure:** $2,810 – $6,760 **Schedule Impact:** 3-5 days

**Recommended Action:** Show refrigerant line set routing on mechanical plans including pipe sizes, insulation requirements, maximum equivalent length verification, and penetration locations through the building envelope with proper sealing details.

Medium

ID: SP029

## Heating System Details Missing from Plans

| | | | |
|---|---|---|---|
| **Disciplines:** | Mechanical ↔ Specifications | **Type:** | Specification Mismatch |
| **Location:** | Spec Page 8 – Central Heating Coil Sizing Data | **Sheets:** | Mechanical plans (spec cross-check) |

**Description:** The specifications indicate a central heating coil load of 23.5 MBH with entering DB of 67.4°F and leaving DB of 78.3°F for AHU2. The heating source type (heat pump, electric resistance, gas) is not explicitly stated in the available spec text, and no mechanical plans exist to show the heating equipment, fuel source connections, or controls. Water flow is listed as N/A, suggesting non-hydronic heating, but the actual system type is undocumented on plans.

| Construct. | Cost | Safety | Schedule | Downstrm. |
|---|---|---|---|---|
| **6**/10 | **5**/10 | **4**/10 | **4**/10 | **5**/10 |

**Cost Exposure:** $2,767 – $6,709 **Schedule Impact:** 3-5 days

**Recommended Action:** Clarify the heating source for AHU2 on the mechanical plans and in the specifications. Show all associated connections (electrical for heat pump/electric heat, or gas piping if applicable) and coordinate with the electrical discipline for circuit sizing.

Medium

ID: C023

## Water Heater Type Mismatch Between Plumbing and Electrical

| | | | |
|---|---|---|---|
| **Disciplines:** | Plumbing ↔ Electrical-Lighting | **Type:** | Specification Mismatch |
| **Location:** | Building-wide | **Sheets:** | P6, E2 |

**Description:** The electrical panel schedule on E2 lists 'WATER HEATER (GAS)' at 0.0 KVA on the demand load calculations for both Panel K and Panel W, indicating the water heater is gas-fired and requires no significant electrical load. The gas isometric on P6 shows gas piping serving various fixtures but the water heater gas connection routing should be verified. The plumbing fixture schedule on P1 shows 'HOT WATER SUPPLY' connections at fixtures. However, the plumbing drawings do not clearly show the water heater location, size, or type on the plan views. Without a clear water heater schedule or equipment callout on the plumbing plans, there is a documentation gap that could lead to field coordination issues regarding the water heater installation, gas connection, venting, and any required electrical connection for controls.

| Construct. | Cost | Safety | Schedule | Downstrm. |
|---|---|---|---|---|
| **3**/10 | **2**/10 | **3**/10 | **2**/10 | **4**/10 |

**Cost Exposure:** $2,558 – $6,688 **Schedule Impact:** 2 days

**Recommended Action:** Plumbing engineer to add water heater to equipment schedule with location, type (gas), capacity (gallons), BTU input, and venting requirements. Show water heater location clearly on plumbing plan. Verify gas line sizing to water heater per P6 gas isometric. Confirm if any electrical connection is needed for electronic ignition/controls.

---

**Medium**

ID: C015

## Plumbing Vent Penetrations Through Roof Not Coordinated with Roof Plan

| | | | |
|---|---|---|---|
| **Disciplines:** | Plumbing ↔ Architectural | **Type:** | Missing Coordination |
| **Location:** | Roof level | **Sheets:** | P4, P5, A5, A4, S1 |

**Description:** The Bathroom Legend on A4 specifies: (A) Two-piece flush-tank 1.28 gallon water closet with elongated seat; (B) Wood-laminated vanity with solid surface countertop and backsplash; (C) Shower stall with frameless glass enclosure, 4" high P.T. curb, waterproof membrane, backer board, marble or tile finish per FBC 2020; (D) Free-standing cast iron bathtub with floor-mounted faucet and supply; (E) Porcelain tile or marble floor. Notes state 'All bathroom fixtures shall be installed as per FBC 2020 (Res.)' and 'All bathroom walls in wet areas shall receive tile/impervious material over Durock cement board to 4x8 minimum height in compliance with FBC Res.' No plumbing drawings are included in this set to verify that waste/vent/supply rough-in locations coordinate with these fixture specifications, particularly the free-standing bathtub requiring floor-mounted supply lines and the shower drain locations.

| Construct. | Cost | Safety | Schedule | Downstrm. |
|---|---|---|---|---|
| **4**/10 | **2**/10 | **2**/10 | **3**/10 | **4**/10 |

**Cost Exposure:** $0 – $0 **Schedule Impact:** 3 days

**Recommended Action:** Plumbing engineer to provide exact coordinates of all roof penetrations to architect. Architect to add all vent penetration locations to the roof plan (A5) and verify they do not conflict with roof drainage slopes, equipment screen, or sundeck areas. Provide flashing details at each penetration per the vent thru roof detail on P4.

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## Medium — ID: C016
## Mechanical Equipment Screen Structural Support Not Detailed

| | | | |
|---|---|---|---|
| **Disciplines:** | Architectural ↔ Structural | **Type:** | Missing Coordination |
| **Location:** | Roof level, equipment screen area | **Sheets:** | M4, M5, S4, A4, A1, S5, A8, A5 |

**Description:** Architectural sections A8 and A1 show an aluminum mechanical equipment screen at the roof level with hurricane louvered screen panels. The notes state: 'THE SCREEN STRUCTURAL FRAMING SHALL BE DESIGNED TO SUPPORT BOTH THE EQUIPMENT SCREEN GRAVITY LOADS AND ALL LIVE LOAD FORCES (IE WIND LOADS) TRANSFERRED THROUGH THE EQUIPMENT SCREEN ASSEMBLY.' Structural sections S4 and S4 note 'SCREEN BY OTHERS PROVIDE SIGNED & SEALED SHOP DWGS FOR REVIEW & APPROVAL BY E.O.R.' The mechanical schedule (M5) shows two condensing units (CU1 and CU2) behind this screen. This is a delegated design item, but the interface between the screen structural framing and the roof slab/parapet requires coordination - the structural drawings do not show any embed plates, anchor bolt patterns, or connection details for the screen attachment to the concrete parapet or roof slab.

| Construct. | Cost | Safety | Schedule | Downstrm. |
|---|---|---|---|---|
| **4**/10 | **3**/10 | **3**/10 | **4**/10 | **5**/10 |

**Cost Exposure:** $0 – $0 **Schedule Impact:** 5 days

**Recommended Action:** Structural engineer to provide embed plates or anchor bolt locations in the parapet/roof slab for screen attachment. Screen fabricator to coordinate connection design with structural engineer. Verify wind load transfer path from screen through parapet to roof structure per the C&C; wind load calculations provided.

## Medium — ID: C017
## Missing Electrical Circuits for Mechanical Equipment - Exhaust Fans

| | | | |
|---|---|---|---|
| **Disciplines:** | Mechanical ↔ Electrical-Lighting | **Type:** | Missing Coordination |
| **Location:** | Second level and roof level | **Sheets:** | E3, M5, M3, E2 |

**Description:** The mechanical equipment schedule on M5 lists exhaust fans EF1,2,3 (Panasonic FV-0511VF1 cabinet fans, 50 CFM, 120V/1/60) and EF4,5,6 (Panasonic FV-0511VF1, 80 CFM, 120V/1/60). The mechanical notes state 'PROVIDE WALL SWITCH FOR ALL TOILET EXHAUST FANS.' However, reviewing the electrical panel schedules on E2, there are no specifically labeled circuits for exhaust fans. The electrical plan E3 shows switches in bathrooms but the panel schedule does not clearly identify dedicated exhaust fan circuits. The mechanical notes also state 'PROVIDE UNIT MOUNTED SPEED CONTROLLER FOR ALL FANS' and 'FURNISH AND INSTALL DISCONNECT SWITCHES AND WIRING FOR VENTILATION SYSTEM FAN MANUFACTURER RECOMMENDATIONS/CONTROLS WILL BE SUPPLIED BY MECHANICAL CONTRACTOR AND CONNECTED BY ELECTRICAL CONTRACTOR' (E3 note 18), indicating a coordination interface.

| Construct. | Cost | Safety | Schedule | Downstrm. |
|---|---|---|---|---|
| **3**/10 | **4**/10 | **3**/10 | **3**/10 | **6**/10 |

**Cost Exposure:** $0 – $0 **Schedule Impact:** 2 days

**Recommended Action:** Verify that exhaust fan circuits are included in the panel schedule, even if combined with bathroom lighting circuits. Confirm that the electrical contractor scope includes wiring from switches to exhaust fans. Add circuit designations to panel schedule for all 6 exhaust fans.

Medium

ID: C018

## Irrigation Pump Missing Electrical Circuit

| | | | |
|---|---|---|---|
| **Disciplines:** | Electrical-Lighting ↔ Plumbing | **Type:** | Missing Coordination |
| **Location:** | Exterior / site | **Sheets:** | E2, A1, P2 |

**Description:** The electrical panel schedule on E2 shows 'IRRIGATION PUMP' listed under Panel 'K' with a circuit (CKT 5, 6) at 20A 2-pole, 1 HP. However, the demand load calculation for Panel 'W' shows 'IRRIGATION PUMP' at 0.0 KVA. There is no irrigation plan or landscape plan provided in this set to confirm the irrigation pump location, size, or power requirements. The site plan (A1) references landscape areas but no irrigation system details are shown. The electrical note on A1 states 'ALL UTILITIES EQUIPMENT AND ACCESSORIES (ELECTRICAL, MECHANICAL, AND PLUMBING) SERVICING THE BUILDING SHALL BE INSTALLED A MIN. OF 12" ABOVE THE BFE (AND 14 IN. PER TAL).' This suggests equipment coordination is needed but irrigation details are absent.

| Construct. | Cost | Safety | Schedule | Downstrm. |
|---|---|---|---|---|
| **3**/10 | **2**/10 | **1**/10 | **3**/10 | **5**/10 |

**Cost Exposure:** $0 – $0 **Schedule Impact:** 5 days

**Recommended Action:** Provide a landscape/irrigation plan showing pump location, GPM requirements, and controller location. Verify the 1 HP pump circuit on Panel K is adequate for the irrigation system demand. Confirm irrigation controller power requirements and add circuit if needed.

Medium

ID: C019

## Emergency Overflow Scupper Locations - Plumbing vs Architectural Coordination

| | | | |
|---|---|---|---|
| **Disciplines:** | Plumbing ↔ Architectural | **Type:** | Missing Coordination |
| **Location:** | Roof level, parapet walls | **Sheets:** | P4, A5 |

**Description:** The plumbing roof level plan (P4) shows 'EMERGENCY OVERFLOW SCUPPER (TYPICAL OF 3) SEE ARCHITECTURAL PLANS' and 'EMERGENCY OVERFLOW SCUPPER (TYPICAL OF 2) SEE ARCHITECTURAL PLANS' at specific parapet locations. The architectural roof plan (A5) shows overflow scupper details and locations, but the count and exact positions need verification between the two disciplines. The plumbing plan references 3 scuppers on one side and 2 on another (total 5), while the architectural plan shows scupper locations that should be cross-referenced. The architectural overflow scupper detail (A5) specifies '8" WIDE x 4" HIGH EMERGENCY OVERFLOW SCUPPER SHALL PARAPET; THE BOTTOM OF SCUPPER SHALL BE A MIN. OF 4" ABOVE THE PRIMARY ROOF DRAIN.' This elevation requirement must be coordinated with the plumbing drainage design.

| Construct. | Cost | Safety | Schedule | Downstrm. |
|---|---|---|---|---|
| **3**/10 | **2**/10 | **2**/10 | **2**/10 | **4**/10 |

**Cost Exposure:** $0 – $0 **Schedule Impact:** 3 days

**Recommended Action:** Cross-reference scupper counts and locations between P4 and A5. Verify scupper invert elevations match between architectural detail and plumbing drainage calculations. Ensure scupper sizing is adequate for secondary drainage per FBC requirements. Confirm scupper discharge locations do not conflict with building facade elements or pedestrian areas below.

Medium

ID: SP034

## Return Air Path Not Documented

| | | | |
|---|---|---|---|
| **Disciplines:** | Mechanical ↔ Specifications | **Type:** | Missing Coordination |
| **Location:** | Spec Pages 8-9 – 2,000 CFM system airflow vs. Mechanical Plans | **Sheets:** | Mechanical plans (spec cross-check) |

**Description:** The specifications indicate a 2,000 CFM constant volume system serving 13 spaces across the second floor. An adequate return air path is essential for system performance, requiring either ducted returns or properly sized transfer grilles/jump ducts between spaces and a central return. The mechanical plans show no return air grille locations, return ductwork, or transfer air provisions.

| Construct. | Cost | Safety | Schedule | Downstrm. |
|---|---|---|---|---|
| **6**/10 | **5**/10 | **4**/10 | **4**/10 | **5**/10 |

**Cost Exposure:** \$0 – \$0 **Schedule Impact:** 3-5 days

**Recommended Action:** Design and document the return air system on the mechanical plans, showing return grille locations, return duct sizing, or transfer grille/jump duct details for each enclosed space. Ensure compliance with code requirements prohibiting return air through bathrooms and laundry rooms.

Medium

ID: SP037

## Window/Skylight Area of 849 SF Not Verifiable on Architectural Plans

| | | | |
|---|---|---|---|
| **Disciplines:** | Architectural ↔ Specifications | **Type:** | Missing Coordination |
| **Location:** | Spec Page 11 - Air System Design Load Summary, Window & Skylight Solar Loads | **Sheets:** | Architectural plans (spec cross-check) |

**Description:** The HVAC design load summary references 849 SF of window and skylight area contributing to solar loads for the second floor zone served by AHU2. This is a substantial glazing area relative to the 1,514 SF floor area (approximately 56% window-to-floor ratio). No architectural window schedule, elevation, or floor plan is available in the plan sheets to verify this glazing quantity. If the architectural window schedule differs, the cooling load calculation of 40.8 MBH sensible could be significantly incorrect.

| Construct. | Cost | Safety | Schedule | Downstrm. |
|---|---|---|---|---|
| **6**/10 | **5**/10 | **4**/10 | **4**/10 | **5**/10 |

**Cost Exposure:** \$0 – \$0 **Schedule Impact:** 3-5 days

**Recommended Action:** Verify the 849 SF window/skylight area against the architectural window schedule and elevations. Confirm glazing specifications (U-value, SHGC) match between architectural specs and HVAC load calculation assumptions.

## Medium — ID: SP036

## Window/Skylight Area Coordination Gap

| | | | |
|---|---|---|---|
| **Disciplines:** | Architectural ↔ Specifications | **Type:** | Missing Coordination |
| **Location:** | Spec Page 11 - Air System Design Load Summary, Window & Skylight Solar Loads: 849 ft² | **Sheets:** | Architectural plans (spec cross-check) |

**Description:** The HVAC design load summary references 849 SF of window and skylight area for the second floor zones served by AHU2. This is a substantial glazing area relative to the 1,514 SF floor area (56% window-to-floor ratio). The architectural plans do not contain readable window schedules or elevations to verify this glazing quantity. If actual installed glazing differs from 849 SF, the cooling load calculation (13,007 BTU/hr solar component) would be significantly affected.

| Construct. | Cost | Safety | Schedule | Downstrm. |
|---|---|---|---|---|
| **6**/10 | **5**/10 | **4**/10 | **4**/10 | **5**/10 |

**Cost Exposure:** $0 – $0 **Schedule Impact:** 3-5 days

**Recommended Action:** Verify the 849 SF window/skylight area against the architectural window schedule and elevations. Confirm glass type, U-value, and SHGC assumptions in the HAP model match the architectural specifications. Recalculate loads if discrepancies are found.

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## Medium — ID: C024

## Frameless Glass Shower Enclosures - Blocking/Support in CMU Walls

| | | | |
|---|---|---|---|
| **Disciplines:** | Architectural ↔ Structural | **Type:** | Missing Coordination |
| **Location:** | Second Level - Master Bath, Bath, and Bedroom #2 W.I.C. area bathrooms | **Sheets:** | A4, S5 |

**Description:** Architectural plan A4 shows multiple 'FRAMELESS GLASS DOOR / ENCLOSURE - USE CATEGORY 1 SAFETY GLASS (TYP.) SHOWER ENCLOSURE' installations in the Master Bath, Bath adjacent to Bedroom #1, and other wet areas. The bathroom legend notes shower stall framing with 2x4 high P.T. curb and waterproofing membrane. The legend on A4 indicates 8" concrete block walls with metal furring channels. Frameless glass shower enclosures require precise blocking or structural support within the wall for hinge and clamp attachments, which must be coordinated with the CMU wall construction. No blocking details are shown on either the architectural or structural drawings.

| Construct. | Cost | Safety | Schedule | Downstrm. |
|---|---|---|---|---|
| **4**/10 | **5**/10 | **7**/10 | **3**/10 | **5**/10 |

**Cost Exposure:** $0 – $0 **Schedule Impact:** 1-2 days

**Recommended Action:** Provide blocking or embed details within the CMU walls at frameless glass shower enclosure attachment points. Coordinate exact hinge and clamp locations with the glass enclosure shop drawings. Include blocking requirements in the structural wall details.

## Medium — ID: SP039

## the structural engineer Firm Than HVAC - Coordination Concern

| | | | |
|---|---|---|---|
| **Disciplines:** | Architectural ↔ Specifications | **Type:** | Missing Coordination |
| **Location:** | Plan Sheets (spBeam output) vs. Spec Pages 9-11 | **Sheets:** | Architectural plans (spec cross-check) |

**Description:** The plan sheet data includes structural beam calculations by the design team Inc. (dated 10/16/2023) using ACI 318-08, while the HVAC load calculations are by the design team (dated 09/18/2023), and architectural sheets are signed by the architect Inc. (dated 10/30/2023). The structural calculations reference ACI 318-08, which has been superseded by ACI 318-14 and ACI 318-19. There is no evidence of cross-discipline coordination notes between these three firms regarding floor-to-floor heights, chase locations for ductwork, or structural accommodations for the 2,000 CFM AHU2 system.

| Construct. | Cost | Safety | Schedule | Downstrm. |
|---|---|---|---|---|
| **6**/10 | **5**/10 | **4**/10 | **4**/10 | **5**/10 |

**Cost Exposure:** $0 – $0 **Schedule Impact:** 3-5 days

**Recommended Action:** Confirm that ACI 318-08 is acceptable under the applicable Florida Building Code edition. Ensure coordination meetings have occurred between the architect, the design team, and the design team regarding duct routing, equipment locations, and structural support for HVAC equipment.

## Medium — ID: C025

## Hanging Rod System and Built-in Millwork Structural Support

| | | | |
|---|---|---|---|
| **Disciplines:** | Architectural ↔ Structural | **Type:** | Missing Coordination |
| **Location:** | Second Level - Master Bedroom closets, Bedroom #2 closet, Laundry | **Sheets:** | A4, S5 |

**Description:** Architectural plan A4 shows multiple 'HANGING ROD SYSTEM W/ UPPER AND LOWER SHELVES' and 'BUILT-IN SHELVING & HANGING ROD SYSTEM (TYPICAL AT CLOSETS) BY OWNER' and 'BUILT-IN MILLWORK' locations throughout the second level. These wall-mounted systems require adequate blocking or support within the 8" CMU walls (per the legend) or the 20 GA. 3-5/8" metal framed interior partitions. The structural notes on S5 do not address blocking requirements for these architectural elements, and the interior partition specification on A4 (20 GA metal studs at 16" O.C.) may not provide adequate support for heavy built-in systems without additional blocking.

| Construct. | Cost | Safety | Schedule | Downstrm. |
|---|---|---|---|---|
| **3**/10 | **3**/10 | **3**/10 | **2**/10 | **4**/10 |

**Cost Exposure:** $0 – $0 **Schedule Impact:** 1 day

**Recommended Action:** Coordinate blocking locations for all built-in millwork and hanging rod systems with the wall framing. For metal stud partitions, specify plywood backing or steel plate blocking at attachment points. For CMU walls, specify toggle bolt or masonry anchor requirements.

# Methodology & Disclaimer

## Analysis Methodology

This conflict analysis was performed using Flikt.AI's proprietary AI-powered plan coordination engine. The system analyzed architectural and engineering plans to identify potential spatial clashes, coordination issues, and conflicts that could impact project delivery. Each conflict has been evaluated across five key dimensions: constructability, cost impact, safety implications, schedule disruption, and downstream effects.

## Confidence Levels

Each identified conflict includes a confidence score (0–1.0) indicating detection reliability. Scores of 0.9+ indicate high certainty; 0.7–0.9 indicate good confidence; below 0.7 warrants manual verification. Human review is recommended for all critical and major severity conflicts.

## Cost & Schedule Estimates

Cost exposure ranges represent estimated rework, delay, and mitigation expenses. Schedule impact estimates indicate typical delay periods based on conflict severity and discipline involvement. Actual costs depend on project-specific factors, contractual arrangements, and resolution strategies.

## Important Disclaimer

Flikt.AI flags discrepancies; resolution is the responsibility of the Architect/Engineer of record. This report represents an automated analysis by Flikt.AI and should not be considered a complete or exhaustive identification of all plan conflicts. The report is intended to supplement, not replace, traditional coordination meetings and detailed plan reviews by qualified professionals. Flikt.AI makes no warranty regarding completeness or accuracy. Project teams should verify all findings before making decisions based on this analysis. Use of this report is subject to Flikt.AI's terms of service.

## About Flikt.AI

**Flikt.AI** — AI-Powered Plan Coordination
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Flikt.AI is the leading AI platform for coordinating complex construction projects, identifying conflicts early, and improving project delivery outcomes.

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